# Linear independence.

① Determine whether the following vector systems in $\mathbb{R}^4$ are linearly indep. or dependent.

a) $V_1 = (1, 2, 2, -1)$
$V_2 = (4, 3, 9, -4)$
$V_3 = (5, 8, 9, -5)$

We have to check if.

$\alpha V_1 + \beta V_2 + \gamma V_3 = 0 \Rightarrow$

$\alpha = \beta = \gamma = 0$ only
(lin. independent case)

Or $\exists\, \alpha, \beta, \gamma$ not all of them $= 0$.
so that $\alpha V_1 + \beta V_2 + \gamma V_3 = 0$.
(lin. independent case).

So

$$\alpha \cdot \begin{pmatrix} 1 \\ 2 \\ 2 \\ -1 \end{pmatrix} + \beta \begin{pmatrix} 4 \\ 3 \\ 9 \\ -4 \end{pmatrix} + \gamma \begin{pmatrix} 5 \\ 8 \\ 9 \\ -5 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\begin{cases} \alpha + 4\beta + 5\gamma = 0 & ① \\ 2\alpha + 3\beta + 8\gamma = 0 & ② \\ 2\alpha + 9\beta + 9\gamma = 0 & ③ \\ -1\alpha + (-4\beta) + (-5\gamma) = 0 & ④ \end{cases}$$

④ · 2.

$-2\alpha - 8\beta - 10\gamma = 0$ ⑤

⑤ + ③

$-5\beta - 2\gamma = 0$

⑤ + ③

$\beta - \gamma = 0$

$$\begin{cases} \alpha + 4\beta + 5\gamma = 0 \\ -5\beta - 2\gamma = 0 \\ \beta - \gamma = 0 \end{cases} \Rightarrow \beta = \gamma$$

$-5\beta - 2\gamma = 0 \Rightarrow -7\beta = 0 \Rightarrow \boxed{\beta = 0}\ \boxed{\gamma = 0}$
$\Rightarrow \boxed{\alpha = 0}$

$\Rightarrow$ Only the trivial linear consultation gives 0 vector
So $V_1, V_2, V_3$ are <u>lin. indep.</u>

b) $V_1 = (1, 2, 3, 1)$
$V_2 = (2, 2, 7, 3)$
$V_3 = (-1, 2, 7, -3)$

Suppose that : $\alpha V_1 + \beta V_2 + \gamma V_3 = 0$.

$$\alpha \cdot \begin{pmatrix} 1 \\ 2 \\ 3 \\ 1 \end{pmatrix} + \beta \begin{pmatrix} 2 \\ 2 \\ 7 \\ 3 \end{pmatrix} + \gamma \begin{pmatrix} -1 \\ 2 \\ 7 \\ -3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\begin{cases} \alpha + 2\beta - \gamma = 0 & ① \\ 2\alpha + 2\beta + 2\gamma = 0 & ② \\ 3\alpha + \beta + 7\gamma = 0 & ③ \\ \alpha + 3\beta - 3\gamma = 0 & ④ \end{cases}$$

① × 2

$2\alpha + 4\beta - 2\gamma = 0$. ⑤

⑤ − ②

$2\beta - 4\gamma = 0$

$\beta - 2\gamma = 0$

$\beta = 2\gamma$ ⑥

$\alpha + 6\gamma - 3\gamma = 0$

$\alpha = -3\gamma$ ⑦

假設:

Suppose $\gamma = 1$:

So. $\beta = 2$.

$\alpha = -3$.

$-3V_1 + 2V_2 + 1 \cdot V_3 = 0$

Coefficients are not all 0.
So $V_1, V_2, V_3$ are <u>linearly dependent.</u>

③ $V_1 = (1, -2, 1) \in \mathbb{R}^3$
$V_2 = (2, 1, 0) \in \mathbb{R}^3$.

Expand this lin. Indep. system by a vector $V_3 \in \mathbb{R}^3$
So that.

a) $V_1, V_2, V_3$ is L.D.

We need $V_3 \in \text{Span}(V_1, V_2)$

For example :

$V_3 := 1 \cdot V_1 + 2 \cdot V_2 =$

$= 1\begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} + 2\begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 5 \\ 0 \\ 1 \end{pmatrix}$

OR.

$V_3 = -V_1 + 3V_2$

$= -\begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} + 3 \cdot \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}$

$= \begin{pmatrix} 5 \\ 5 \\ -1 \end{pmatrix}$

b) $V_1, V_2, V_3$ is L.I.

So we need $V_3 \notin \text{Span}(V_1, V_2)$

So. $\text{Span}(V_1, V_2) =$

$= \left\{ \alpha \cdot \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} + \beta \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} \,\middle|\, \alpha, \beta \in \mathbb{R} \right\}$

$= \left\{ \begin{pmatrix} \alpha + 2\beta \\ -2\alpha + \beta \\ \alpha \end{pmatrix} \in \mathbb{R}^3 \,\middle|\, \alpha, \beta \in \mathbb{R} \right\}$

If $\alpha = 1 \Rightarrow$

$\begin{pmatrix} 1 + 2\beta \\ -2 + \beta \\ 1 \end{pmatrix}$ and.

$V_3 = \begin{pmatrix} 1 + 2 \cdot \beta \\ -2 + \beta \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ -2 \\ 1 \end{pmatrix}$

<u>or</u>

$V_3 = \begin{pmatrix} 1 + 2\beta \\ -2 + \beta \\ 2 \end{pmatrix}$ $\beta \in \mathbb{R}$ is also good

$\beta = 1 \Rightarrow V_3 = \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix}$